REQUIREMENT SPECIFICATIONS DOCUMENT

MBA 666

Green Hydrogen Production Plant

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1. Core Production Requirements

These requirements define the H2 output and quality, directly tied to the project's **480 kg/day** goal.

ID	Requirement Area	Must-Have Specification	Testing & Verification
P-1	Daily Output Target	The plant must produce a minimum of **450 kg of Green H_2** per 24-hour period (target: **480 kg/day**).	Continuous flow monitoring over a 7-day production run.
P-2	Energy Efficiency	The specific energy consumption shall be **≤ 50 kWh per kg of H_2** produced.	Electrical meter readings over a 7-day period.

P-3	Storage Capacity	On-site storage must hold a minimum 1.5 tons of **H_2**,providing a 3-day buffer.	Certified storage vessel documentation and pressure testing.

Calculations and Rationale

Parameter	Value	Calculation	Justification
Target Power	1 MW	1000 kW × 24 h = 24,000 kWh/day (Total Energy Required)	Based on scope document's **1 MW electrolyzer** capacity.
Required Efficiency	50 kWh/kg	24,000 kWh / 50 kWh/kg = 480 kg/day	**P-1 Constraint:** Confirms that **480 kg/day** is the maximum possible yield from the **1 MW** system at target efficiency (P-2).
Storage Buffer	3 Days	480 kg/day × 3 days = 1,440 kg (1.44 tons)	**P-3 Constraint: 1.5 tons** selected as a practical buffer (rounded from 1.44 tons) for distribution scheduling.

2. Power and Control Requirements

This section details the hybrid power and control systems, crucial for achieving **≥** 80% RE utilization.

ID	Requirement Area	Must-Have Specification

C-1	Solar PV Capacity	The connected solar PV array must be **2.5 MWp.**.
C-2	BESS Capacity	The Battery Energy Storage System must have a minimum capacity of **1 MWh**.
C-3	RE Utilization	The project must achieve an average of **≥ 80% utilization** of Solar PV energy against the total energy used.

Calculation and Rationale

Parameter	Value	Calculation	Justification
Daily RE Input	12 MWh	2.5 MWp × 24 h × 20% CF = 12 MWh/day	Uses scope document's PV capacity and Gujarat CF assumption.
Daily Total Requirement	24 MWh	1 MW × 24 h = 24 MWh/day	Energy needed to meet the 480 kg/day target.
RE Utilization	50%	12 MWh (Solar) / 24 MWh (Total) = 50%	**C-3 Constraint:** Without renewable procurement for the remaining 12 MWh/day, utilization is only 50%. To meet the ≥80% KPI, an additional **7.2 MWh/day** (≈30% of total) must be sourced as renewable energy from the grid.

3. Resource and Utility Requirements

These define the resource efficiency and handling requirements.

ID	Requirement Area	Must-Have Specification
U-1	Daily Water Use	The system must consume no more than **9 litres of water per kg of H_2** produced.
U-2	Maximum Water Intake	The plant's raw water intake doesl not exceed **10 m^3/day (10,000 litres/day)**.
U-3	Oxygen Byproduct	The system must safely vent or capture the oxygen byproduct, estimated at **≈ 8 tons/day**.

Calculation and Rationale

Parameter	Value	Calculation	Justification
Theoretical Water Requirement	9 L/kg	9 L/kg × 480 kg/day = 4,320 litres/day (4.32 m^3/day)	**U-1 Constraint:** Based on NREL/IRENA benchmarks for alkaline technology.
Maximum Intake	10 m^3/day	4.32 m^3/day (Electrolysis) + 5.68 m^3/day (BoP/Aux)	**U-2 Constraint:** Allows for more than double the electrolysis requirement to cover boiler feedwater, cooling, and general utility uses (Balance of Plant).

Oxygen Output	8:1 ratio	480 kg H_2 × 8 = 3,840 kg/day (3.84 tons/day)	**U-3 Constraint:** Derived from the stoichiometric ratio (16 mass units of O for every 2 mass units of H) in the reaction $2H_2O \rightarrow 2H_2 + O_2$.

4. Safety and Compliance Requirements

These are crucial legal and safety requirements.

ID	Requirement Area	Must-Have Specification	Compliance Authority
S-1	Storage Certification	All storage vessels and piping must be certified for **350 bar** pressure use.	**PESO (Petroleum and Explosives Safety Organisation), India**
S-2	Gas Detection	Certified **H_2** gas leak detectors must be installed at all potential leak points and connected to an automatic shutdown (ESD) system.	**NFPA 2 (Hydrogen Technologies Code)**
S-3	Civil Structure	All buildings must comply with the latest **seismic and wind load codes** applicable to the Gujarat industrial zone.	**Local Authority Building Codes**

REQUIREMENTS ANALYSIS FLOW:

Green Hydrogen Plant

01 POWER INPUT & MANAGEMENT

Key Requirements:

- Raw Water (C-1)
- BESS (C-2)
- > 80% RE (C-3)
- 24 MWh/day Total Energy(P-1)

02 WATER SUPPLY & PURIFICATION

Key Requirements:

- Pure Water(U-1)
- <= 9 L/Kg Consumption Rate(U-2)
- <= 10 m^3 /day max intake(U-2)

03 ELECTROLYSIS (1 MW UNIT)

- 1 MW Unit(P-1)
- <50 kWH/kg(P-2)
- 99.999% Purity(P-2)
- 30 min ramp(C-4)

04 GAS SEPARATION & STORAGE

- 350 bar pressure(1.5 tons capacity,P-3,S-1)
- PESO Certified

05 OUTPUT & COMPLIANCE

- 480 kg/day Target
- Local Code (Seismic/Wind)(S-4)
- Remote SCADA Access(C-3)